Visualizing Trends and Affordability in the Canadian Housing Market

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Abstract

In recent years, housing prices across Canada have climbed significantly, leading to growing concerns about affordability in several regions. This project set out to explore how prices have been changing by examining housing data from Kaggle alongside benchmark figures provided by the Canadian Real Estate Association (CREA) and Canada Mortgage and Housing Corporation (CMHC) and from Canadian Housing Statistics Program (CHSP). We used Power BI to build interactive dashboards that include line and bar charts, along with slicers, to track price trends for different types of homes between 2020 and 2025. The tool lets users filter by province, housing type, and year to explore how prices have moved and how different regions compare. One of the main conclusions drawn from our analysis is that price growth hasn't been uniform across the board; in certain provinces, for instance, two-story homes have seen particularly significant increases. Overall, the visuals make it easier to spot patterns and provide useful insights for buyers, policymakers, and anyone interested in the housing market.

Keywords: Canadian housing market, affordability, Power BI, data visualization, exploratory data analysis, CREA, benchmark prices, interactive dashboard, real estate trends

I. INTRODUCTION

Over the past few years, housing prices in Canada have gone up quickly, making it harder for many people especially in provinces like Ontario and British Columbia—to afford a home. Although national and provincial reports offer summaries of housing data, they often don't provide the flexibility or interactivity needed to dig deeper into trends by region or property type.

To help fill that gap, this project uses Power BI to build an interactive dashboard that lets users explore benchmark housing price trends from 2020 to 2025. Using data from Kaggle and the Canadian Real Estate Association (CREA), we look at how prices differ across provinces and between property types, like apartments and two-storey homes. The main goal is to create a user-friendly tool that makes it easier to understand housing market trends whether you're a data analyst or just someone interested in real estate.

The rest of the report goes over past research, explains the data and methods we used, highlights the key findings through visualizations, and ends with a summary of what we learned and possible next steps.

II. PREVIOUS WORK

Housing affordability has become a major topic in Canada, drawing increased attention from both government bodies and real estate groups. The Canadian Real Estate Association (CREA), for instance, provides the MLS® Home Price Index (HPI), which adjusts for variations in housing types to give a clearer picture of price trends over time [1]. Similarly, the Canada Mortgage and Housing Corporation (CMHC) regularly releases Housing Market Outlooks that offer forecasts and regional insights into market conditions [2]. Statistics Canada adds to this with detailed ownership and property info through its Canadian Housing Statistics Program [3].

Even though these sources are reliable and offer a lot of data, most of it comes in the form of static reports or spreadsheets. That makes it harder for people to explore the data based on what they care about, like specific regions, timeframes, or types of homes. On the other hand, community platforms like Kaggle have also added some helpful datasets (like Home Prices in Canada 2021–2022 [4] or Canada Housing Benchmark [5]), but using them usually involves coding and tools that aren't very beginner-friendly.

This project tries to bridge that gap. We took CREA's benchmark price data and combined it with detailed info from Kaggle, then built an interactive dashboard in Power BI. The idea is to give people a tool they can use to dig into price trends by province, by year, or by housing type without needing to know how to code.

III. METHODOLOGY

A. Data Sources and Scope

Two datasets were used to study housing trends in Canada. The first one is the MLS® Home Price Index (HPI), shared by the Canadian Real Estate Association (CREA). It includes benchmark prices for different types of homes, like apartments, townhouses, and detached houses across various provinces and cities. The benchmark prices are adjusted for seasonal effects to account for regular patterns, like higher activity in the summer months. This table helps give a clearer view when comparing prices across different years. The second dataset,taken from Kaggle and titled Canada Housing, includes information on individual homes, such as the number of bedrooms, bathrooms, and square footage. Instead of merging it with the CREA data, it was used separately to create a dashboard that explores how these property features influence pricing.

B. Data Cleaning and Preprocessing

The data from the CREA dataset was cleaned and prepared in Power BI using Power Query. Several tables, including those for cities, provinces, house types, and benchmark prices, were first imported and then joined to create two main datasets: one for city-level analysis and another for provincial trends. To make the data usable, price values were converted from text to decimal format, and date fields were formatted correctly. Records before 2020 were excluded to keep the focus on the most recent five years. All rows that contained only null values were removed. To make the data easier to work with in visualizations, the housing-type benchmark columns were unpivoted, resulting in a long format with separate “House Type” and “Price” columns.

C. Data Modeling and Measures

We also did exploratory data analysis (EDA) on the Kaggle dataset to learn more about how different property-level features affect housing prices. We looked at things like bedrooms, bathrooms, square footage, and price using correlation heatmaps, scatter plots, and box plots. The analysis noticed a strong linear relationship between the square footage and price, as well as moderate links between the number of bedrooms and bathrooms. Even though latitude and longitude showed low correlation values to price, they were kept for possible geospatial analysis. Python was used to preprocess the Kaggle dataset, which included cleaning the data, dealing with missing values, and selecting the right columns based on EDA insights. The dataset also had a lot of outliers in variables like price and square footage, so median values were used instead of mean values in visualizations to get more reliable and representative insights. DAX was only used in the CREA dataset, not Kaggle, for dynamic calculations. For example, it was used to temporarily change the format of certain columns to a longer one without changing the whole table. We also used DAX variables to change the names of fields in expressions to make the visuals clearer. DAX also made a field parameter called "House Type" that lets users switch between different types of homes, like one-story, two-story, apartment, townhouse, and single-family, all in the same visual component. This made the dashboard a lot more flexible, clear, and interactive.

D. Dashboard Design and Visual Elements

Two dashboards were created based on the CREA dataset: one for cities and one for provinces. Each dashboard includes a clustered bar chart showing house-type prices for each year, overlaid with a median line for comparison. A 100% stacked bar chart displays yearly percentage change in benchmark prices. KPI cards highlight the exact median values of apartments and composite housing. The province dashboard also includes a map visualization to show spatial differences in housing prices across Canada. Users can interact with the dashboards using slicers and filters for city, province, and house type to explore different comparisons and trends.

A third dashboard was created using the Kaggle dataset, which focuses on how property-level features such as bedrooms, bathrooms, and areas (in square feet) affect housing prices. This dashboard includes scatter plots, box plots, and distribution charts to visualize pricing patterns based on different configurations.

E. Interactivity and Usability

The dashboard is fully interactive; when a user clicks on a chart or picks something from a slicer, the other visuals update right away. This makes it easy to compare prices across different provinces, cities, or types of housing. Since the visuals are all tied together through the data model, everything responds in real time. All the steps for cleaning and calculating the data were saved in the Power BI file, so the whole setup can be refreshed later if new data is added. It also means the same dashboard can be reused for future analysis without starting from scratch.

V. RESULTS

These interactive dashboards let users explore the Canadian housing market in depth and in a way that makes sense, giving them useful information from many different angles. The first dashboard demonstrates trends at the provincial level. It shows the median house prices over the years, comparisons between different types of houses, and a forecast of benchmark prices for the future. This lets users see patterns from the past and make predictions about changes in the future. The second dashboard goes deeper into city-level data by showing the median prices for different types of homes year after year. This feature makes it easier to compare prices and see how they change over time in different cities. The third dashboard is an advanced analytical tool that lets you filter by number of bedrooms, bathrooms, parking availability, and location. It has a dynamic median price card, an interactive map to show how prices are spread across regions, and a scatter plot that shows how square footage and price are related. This analysis considers taking into account the number of bedrooms. These dashboards supply users—from analysts to homebuyers—a complete, data-driven picture of housing trends. They help people make smart choices based on real estate patterns, price changes, and differences between regions in Canada.

A. Provincial Housing and Forecasting Insights

Figure 1: Dashboard For National Housing Price Trends (2020–2025)

This dashboard gives the user a full picture of Canada's median house prices from 2020 to 2025. Some of the most important points are:
Year-over-Year Growth: The cbar chart in the top left shows that median house prices have gone up every year for six years, from $315.15K in 2020 to $467.45K in 2025 (a 48.3% rise). This type of graph provides an insight about the changes in price across years.House Type Breakdown: The card visuals show that prices vary by type of housing. Single-family
Homes are the most expensive ($407.45K), while apartments are the least expensive ($294.75K). This type of slicer provides the user the ability to filter data based on their interests.

B. Provincial Housing and Forecasting Insights

Figure 2. Dashboard For National Housing Price Trends (2020–2025)

This dashboard focuses on the Ontario province and summarizes trends for apartments, single family, and townhouses.
The trend in the Ontario province showed that prices increased from $670.75K in 2020 to $929.05K in 2022, then gradually decreased to $807.95K in 2025, suggesting a market correction, with single-family homes reaching a peak price of $940.25K, which was much higher than that of townhouses ($682.60K) and apartments ($580.95K).
Comparative house types and forecasting: The price difference is clear, with apartments displaying relative affordability and single-family homes holding the highest value. The forecast predicts that the market will begin to recover in 2026 with the goal of reaching new price peaks by 2028.

C. City-Level Median Price Comparison

Figure 3. Dashboard For City-Level Median Price Comparison (Comparison of house price in Toronto with One-Story, apartment, and Townhouse)

This dashboard shows housing trends in major cities. The horizontal bar chart on the left shows how the national median price has steadily gone up from $484.25K in 2020 to $623.35K in 2025, with a small level drop seen between 2023 and 2024. Interactive filters that let users see how prices change by city and house type add even more to the regional insights. These filters help to explain how housing works in cities. The dashboard shows that median prices in Toronto peaked at $1.12 million in 2022 and then slowly dropped to $0.99 million in 2025. A column chart that compares prices shows that one Storey houses have the highest prices, while apartments and townhouses have prices that stay pretty stable. A big number card shows that the median price for Toronto (visible city wise upon selection) house types is $1.04 million, which shows how expensive the city is for housing. Users can also look at price trends in their area over time with customizable filters.

D. A close-up look at the Toronto real estate market, with insights at the micro level

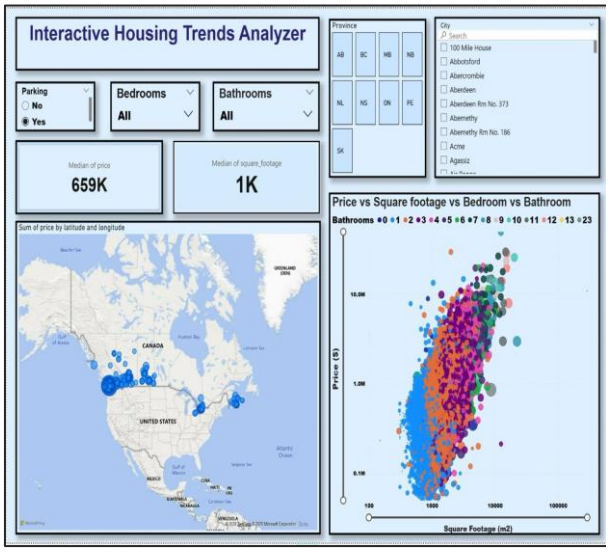

Figure 4. Dashboard For Interactive Housing Trends

With this interactive dashboard, users can have a look at the housing trends across Canada using filters. The Geospatial Information has the map visual, which shows groups of property prices in different provinces and cities, highlighting regional concentration.

Price vs. Square Footage Analysis helps the user know using The scatter plot to show the relationship between square footage and price, with colors showing the number of bathrooms. This helps find pricing anomalies or outliers.

The Users can use dropdowns to choose the number of bedrooms, bathrooms, parking spaces, provinces, and cities to explore using the filters obtained using slicers.

The Key stats like the national median price ($659K) and the average square footage (1,000 sq. ft) gives the user quick information.

E. A close-up look at the Toronto real estate market, with insights at the micro level:

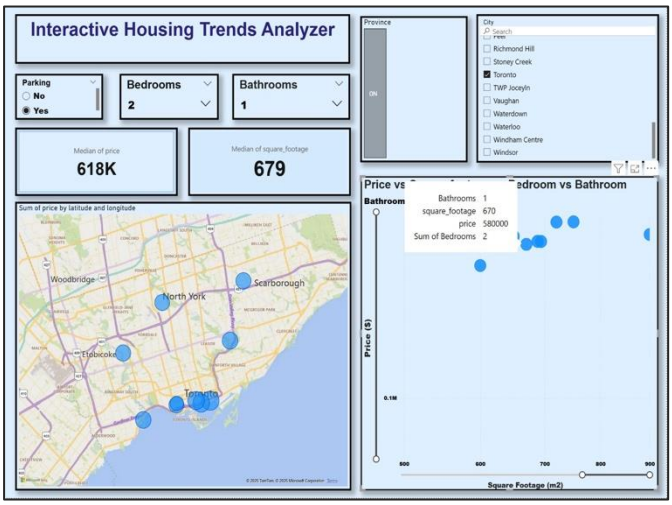

Figure 5. Dashboard For Toronto houses based on house configurations (square footage, bedrooms, bathrooms, and parking).

Localized Mapping: The map shows where property prices are highest in Toronto and nearby boroughs like Scarborough, North York, and Etobicoke.
Specific Listing Breakdown: The scatter plot's tooltip shows detailed listing information, such as the price ($580K), square footage (670), number of bedrooms (2), and number of bathrooms (1).
User-Driven Filters: Sliders and checkboxes let users narrow down their choices, making it easier to compare options that are right for them.
Toronto Median Stats: The median price of $618,000 for a 679-square-foot unit shows how affordable 2-bedroom apartments are in the city

V. CONCLUSION

The Canada Housing Trends Analysis Project gives an interactive view of how real estate has changed across the country from 2020 to 2025.

We have used Power BI to make dashboards that show housing data for the whole country, each province, and each city. They also show how prices have changed over time for different types of properties and make predictions about what will happen in the future to help people make wise choices.

The analysis shows that median house prices are steadily going up at the national level, going from about $315,000 in 2020 to more than $467,000 by 2025.

This trend shows that the housing market is growing quickly because of the recovery from the pandemic, higher demand, and a lack of housing. When comparing different types of homes, single-family and two-story homes have the highest prices. Apartments, on the other hand, are the most affordable option, making them easier for first-time buyers to get into.

The forecast part adds predictive value by showing that housing prices are expected to go back up in the years after 2025, even though there will be a temporary drop. These insights can help investors, real estate agents, and policymakers get ready for short-term changes while keeping an eye on long-term value.

The interactive dashboards made the analysis easier by giving users the ability to discover homes based on their requirements. Users can look at price differences based on square footage, number of bedrooms, bathrooms, and parking availability with tools like filterable slicers, geospatial maps, and scatter plots.

Basically, the dashboards do a good job of showing how prices change over time, in different areas, and for different types of properties. They not only show historical trends, but they also give stakeholders the tools they need to find market opportunities, understand gaps in affordability, and plan future investments. This project shows how powerful Power BI is at turning raw housing data into clear, useful information that helps people make better decisions and plan better for real estate.

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